A Capability Proof, by Construction:

Automated theorem-proving on a qualifying-exam corpus, with graded warrants and machine-checkable certificates

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Abstract

We are formalizing and solving a corpus of ∼493 mathematics qualifying-exam problems ("APM") in Lean 4/Mathlib, using a relay of LLM agents. Rather than reporting a solve rate, we organize the whole programme as a *proof by construction* of the claim "all APM problems are solved": the constructive content is an automated pipeline; the claim decomposes into sub-claims (typed holes with contracts); and each sub-claim carries a *graded warrant* backed by machine-checkable certificates — compiler exits, axiom audits, statement hashes, append-only ledgers — rather than narrative. The pipeline closed its first theorem end-to-end under automation (Steinhaus's theorem) on the day the driver was built. We describe the method, the current warrant state (including the deliberately weak nodes), and the causal-inference layer used to keep claims about *why* components work identified rather than merely correlated. The intended contribution is less the solve rate than the epistemic form: a research programme whose own capability claim is held to the same standard of proof as its object-level theorems.

1 The claim, read constructively

The headline claim — *all APM problems are solved* — is false today (∼27% of the corpus is closed) and will not be proved by enumeration. Read constructively (in the Brouwer–Heyting–Kolmogorov sense), the claim asserts instead that a *procedure* exists which, handed any APM problem, produces a mechanically-verified solution — together with witnesses that each component of the procedure does its job. The proof object is the pipeline; the corpus percentage is merely its odometer.

This reading changes what counts as progress. Evidence attaches to *components* (sub-claims), each stated with a contract and a current warrant; the top-level claim is discharged when the component warrants are strong and the composition runs continuously. The document that holds this structure is a live file, updated mechanically by the pipeline itself, under one rule: **a warrant upgrades only by certificate, never by narrative.**

2 Warrants and certificates

Six warrant classes are in use: *mechanical* (an executed witness: the Lean compiler exits 0, the axiom audit is clean, a statement hash matches); *replicated* (independent runs agree on

*Prepared with the futon agent mesh (Claude, Codex, and Z.ai seats coordinated over a local Agency); the human's role was direction, review checkpoints, and the judgments marked as such below.

the load-bearing quantities); *inductive-*n*=*K (K observed instances, honesty bounds stated); *designed* (contract written, no witnesses yet); *registered* (candidate mechanism, not yet applied); and *refused* — which distinguishes *proved-impossible* from *not-yet-capable*, because conflating those two species of "no" destroys planning.

Certificates are cheap and adversarial by construction: every code module was built by one agent and reviewed by another (author $\neq$ reviewer, enforced in the memory store's attachment lifecycle as well as in process); every claimed number was re-derived by the reviewer from the artifacts; mock-confirmed assumptions are explicitly distrusted after two same-day incidents in which a test suite faithfully validated its author's invention while the live system disagreed. Heterogeneous warrants are the point, not a defect: an honest proof marks precisely where certification is absent.

3 The construction

The pipeline is a relay with a division of labour that emerged from measurement, not doctrine:

1. **Starter** (Z.ai seat): reads the informal problem (TeX + prose), performs a memory-reconnaissance pass over the programme's evidence store, then formalizes a faithful Lean statement and proves as far as it honestly can. A partial ends at a *boundary artifact*: the smallest possible `sorry` at the exact missing bridge, with a comment recording APIs searched, routes tried, and routes *not* investigated.
2. **Mechanical gate** (no model in the loop): build, a comment-stripped `sorry` count, a verbatim `#print axioms` audit, a whitespace-stable statement hash (frozen at first formalization; any later mismatch voids the chain), and a boundary-conformance check.
3. **Closer** (Codex seat, ≤ 3 hops): receives the boundary artifact and closes the hole — by finding the missing lemma, proving it locally, or routing around it. Desk research (Mathlib source, previously solved neighbours, git history) is framed as part of the job, with discards recorded with reasons.
4. **Human-judgment checkpoints, and only these**: fidelity of a fresh formal statement to its informal source (adjudicated against the TeX contract), and anomalies. Both arrive as structured requests with everything needed to adjudicate; verdicts are files; everything else is automatic.
5. **Scribe**: after each chain, an extraction pass distills reviewed memories (lemma locations, reusable proof patterns, consultation practices) into the evidence store, including a *hunger audit*: retrieval queries that returned nothing are recorded as demand signals, and memories answering them carry the asker's literal vocabulary as tags.

Every transition is a record in an append-only ledger under a registered identity; state is derived by folding the ledger, illegal transitions are unrepresentable, and the operator can inspect any chain at any moment. These properties are requirements extracted from a failure (an earlier opaque cron dispatcher was retired for exactly this), not aspirations.

4 Current warrant state

Sub-claim	Warrant	Key certificates
N1 Extra resources fill Mathlib holes	inductive-n=3	interval-decomposition lemma proved locally in ∼15 min; Liouville derived without the packaged theorem; L^1 translation-continuity bridge found and used (Steinhaus)
N2 Work transports between agents	inductive-n=4	boundary artifacts paved each closer's path; cross-problem reuse cited in source; protocol itself stored as a memory
N3 The store records learning	inductive-n=4 sessions	20 reviewed, tagged, attached memories in two days; extraction quality held without correction; hunger audit ran unattended
N4 Agents consult memory when instructed	inductive-n=1 (controlled)	same agent, same store: 0 lookups under a passive invitation vs. 21 under a two-part task frame
N5 Retrieval serves the need	**weak**	a four-layer failure anatomy (propensity / framing / affordance / index-reach); the one candidate use-witness is *cited-only* — the memory is cited in a proof body, but the citing theorem does not compile (12 errors under an executed check), so no step or problem is yet witnessed as closed by a retrieved memory. Detail lives in the sibling store proof (node M4)
N6 Capability transports to held-out problems	designed	stated as a formal transportability claim; no witnesses yet
N7 Outcomes are mechanically scoreable	designed→mechanical	delta-form endpoint, executed-witness-only scoring; capture landing
N8 The process learns at the ability level	designed	lessons travel via versioned packet templates; automation of template deltas not yet built
N9 The pipeline runs continuously	inductive-n=2 chains	driver built in one day from six independently reviewed modules; supervised trial witnessed both branches (see below)

The two trial chains are worth a sentence each, because together they witness the composition. **Chain 1** (Dirichlet integral): the starter session died mid-work at 43 minutes; the gate correctly refused the corpse; the anomaly checkpoint fired with a complete dossier; the human verdict (*abandon*) folded on restart; two instrument defects were found and fixed as a result. The safety branch works, and failures are data. **Chain 2** (Steinhaus's theorem): the starter produced an exemplary honest partial — the exact missing bridge documented with APIs searched and routes untried; a gate false-positive was caught at the checkpoint, the *instrument* was fixed rather than the artifact, and on resume the closer supplied the missing continuity bridge (`tendsto_measure_symmDiff_preimage_nhds_zero`), closing the theorem: build clean, zero `sorry`s, axioms exactly `[propext, Classical.choice, Quot.sound]`, fidelity adjudicated

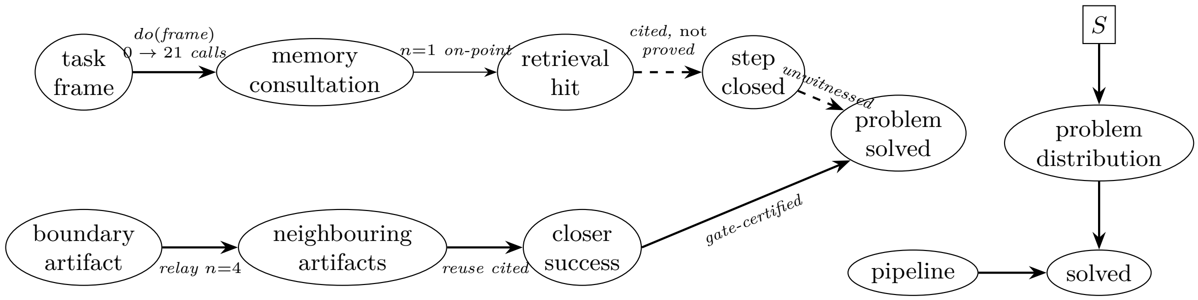

Figure 1: **Left:** the mechanism DAG as currently witnessed. Solid edges carry certificates (labels). The retrieval edge is split by granularity, and *both halves remain dashed.* A candidate step-level witness (a96A08) was examined on 2026-08-04: the runner did cite an on-point memory in the proof body immediately above the lemma application — so the citation is real, and the earlier claim that every hint was discarded as off-route no longer holds — but the enclosing theorem **does not compile**. An executed `lake env lean` reports 12 errors, the first at line 136, inside the cited theorem itself. The claim that it was proved was withdrawn the same day: it rested on reading the tail of a truncated compiler log, which showed only the last two of twelve diagnostics. Citation without a compiling artifact is a *cited-only* witness, not a use-witness. The $do(\cdot)$ edge is the programme's one true intervention: a preregistered controlled contrast on the task frame. The lower path is the relay: boundary artifacts and neighbouring solved problems feed closer success (n=4 witnessed hops), which the mechanical gate certifies into solutions. **Right:** the transport question N6 as a selection diagram: S switches the problem distribution (APM vs. held-out BPM); "the pipeline works on BPM" is identified only if the S-dependence can be separated from the pipeline edge — a derivation obligation, not an extrapolation.

against the TeX contract. The scribe then wrote four memories — including the programme's first *open-hunger* record: a query the corpus could not answer, banked as demand.

5 The causal layer

"The memory system helps" is a causal claim, and we treat it with the same discipline as the theorems (Figure 1). Contrasts are designed before they are run (the n=1 controlled contrast in N4 was preregistered, with its predictions written down and two of them subsequently refuted by the data — refuted predictions being exactly what preregistration is for). Identification precedes estimation: a small causal engine (Pearl-style: backdoor and front-door adjustment, general identification, transportability via selection diagrams) supplies receipts for which contrasts are identified by design and refuses the rest; N6 above is stated in its vocabulary because "solves APM $\Rightarrow$ capable on held-out BPM" is formally a transport claim, not an induction. Where evidence is observational or n is small, the warrant says so; nothing in the table above launders correlation into mechanism.

6 Honest boundaries

N5 is weak and is named as the binding constraint on the memory loop: agents now reliably *ask* (N4), but the store does not yet reliably *answer*, for reasons diagnosed to the index and affordance layers, with mechanical repairs queued. N6 has no witnesses. The corpus is $\sim$27% closed and the inductive n's are small everywhere — the table is a statement of what has been witnessed, not of what is expected. Two structural findings temper optimism about memory in particular: within a relay chain, the *repository itself* (boundary comments, neighbouring solved problems) has so far been the memory channel that demonstrably works; the store's value is amortized *across* chains and is only now becoming measurable. And instructions dominate architecture at every margin measured so far: one plain sentence of task framing moved memory-tool usage from zero to twenty-one calls.

7 What this is for

The pipeline will keep eating the corpus, and the live document will keep upgrading warrants by certificate. But the transferable object, we think, is the form: a research programme that states its own capability claim as a constructive theorem, decomposes it into contracted holes, refuses to let anything upgrade without a machine-checkable witness, and keeps its refusals typed. The same form is now being applied, by a second agent lane, to a harder subject: the memory architecture itself.